

TUSCOLA AREA AP, MI, USA

WMO: 722007

Lat: 43.459N

Lon: 83.446W

Elev: 214

StdP: 98.78

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54828

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-16.4	-23.7	0.5	-19.3	-20.6	0.6	-15.4	10.8	-4.8	9.8	-2.9	1.7	300	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	32.0	22.4	30.2	21.6	28.6	21.1	24.1	28.8	23.1	27.9	22.3	26.7	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	17.8	26.8	21.9	17.0	25.6	21.0	16.0	24.8	73.3	29.0	69.7	27.8	66.6	26.5	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.3	7.4	DB	-24.5	34.5	4.3	1.6	-27.5	35.6	-30.1	36.6	-32.5	37.5	-35.6	38.7	
			WB	-24.7	25.9	4.2	1.1	-27.8	26.7	-30.2	27.3	-32.6	27.9	-35.6	28.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-5.1	-4.7	1.0	7.3	14.2	18.9	21.3	20.3	16.4	10.3	3.7	-1.5
	DBStd	10.57	6.37	6.38	6.50	4.91	4.93	3.79	3.29	2.94	4.20	4.84	4.99	5.09
	HDD10.0	1906	468	412	288	109	14	0	0	0	2	57	198	357
	HDD18.3	3844	726	645	538	331	149	38	10	14	85	254	440	614
	CDD10.0	1387	0	0	9	29	144	267	349	318	196	66	8	1
	CDD18.3	284	0	0	1	1	21	54	101	74	28	4	0	0
	CDH23.3	2894	0	0	7	19	257	589	1050	670	262	39	0	0
	CDH26.7	837	0	0	1	2	64	181	346	171	67	6	0	0
Wind	WSAvg	2.9	3.6	3.6	3.4	3.6	2.8	2.4	2.0	1.9	2.1	2.9	3.4	3.6
Precipitation	PrecAvg	841	54	46	50	77	89	92	88	74	76	78	69	54
	PrecMax	1052	101	108	103	158	185	160	146	106	155	159	140	106
	PrecMin	719	13	15	15	30	42	33	16	34	13	21	16	14
	PrecStd	102	24	25	24	34	36	35	35	22	40	33	33	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.1	23.7	26.2	31.0	32.8	34.1	32.5	32.0	27.6	20.0	14.3
		MCWB	10.3	10.7	16.5	15.1	20.9	22.4	24.2	22.2	21.4	21.2	13.7	12.5
	2%	DB	8.3	9.8	17.8	22.5	28.4	31.1	32.2	30.2	28.5	24.0	16.4	10.2
		MCWB	6.6	7.7	12.8	13.2	19.5	21.5	23.1	21.9	21.4	17.3	11.5	8.4
	5%	DB	5.6	6.0	14.0	19.9	26.6	28.7	30.2	28.4	26.3	21.3	13.9	7.5
		MCWB	3.9	3.8	10.0	12.6	18.5	20.8	21.9	21.4	19.8	16.7	10.2	5.5
	10%	DB	3.0	3.0	10.9	17.0	23.9	27.0	28.4	27.1	24.3	18.7	11.9	5.1
		MCWB	1.8	1.3	7.4	10.7	16.4	19.3	21.3	20.4	18.5	15.1	9.0	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	11.8	16.8	17.6	22.4	24.1	25.7	25.1	24.4	21.7	15.5	12.8
		MCDB	11.6	13.4	23.5	23.2	28.6	29.2	31.4	30.1	29.0	26.4	18.1	14.2
	2%	WB	7.0	7.7	13.8	15.2	21.0	22.9	24.4	23.6	22.5	19.6	12.7	8.8
		MCDB	8.1	9.7	17.2	20.0	26.2	28.5	29.5	27.6	26.5	22.6	15.4	10.1
	5%	WB	4.1	4.1	10.8	13.3	19.7	21.9	23.4	22.6	21.1	17.3	10.6	5.9
		MCDB	5.5	5.8	14.5	18.3	24.4	26.6	28.2	26.6	24.3	19.7	13.4	7.3
	10%	WB	2.2	1.7	7.3	11.2	18.0	20.8	22.5	21.6	19.9	15.3	8.8	3.4
		MCDB	3.3	3.2	10.2	16.6	22.6	25.0	26.9	25.4	23.2	18.1	11.6	4.9
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.8	9.8	11.6	12.5	12.6	12.5	11.9	12.2	10.2	8.5	6.3
		MCDBR	9.2	10.0	14.6	16.5	15.1	14.4	14.1	13.4	13.9	13.9	12.2	9.4
	5% WB	MCWBR	8.0	8.4	9.8	9.2	7.9	7.1	6.5	6.5	7.2	8.1	8.6	8.0
		MCWBR	8.8	9.6	13.6	14.1	13.4	12.1	11.9	11.2	11.8	12.0	10.8	9.2
Clear-Sky Solar Irradiance	taub	taub	0.291	0.309	0.334	0.367	0.392	0.419	0.410	0.402	0.373	0.347	0.310	0.286
		taud	2.383	2.328	2.373	2.307	2.292	2.244	2.289	2.300	2.376	2.437	2.496	2.441
	Ebn at Noon	Ebn at Noon	842	884	906	897	880	854	858	853	852	831	817	816
		Edh at Noon	77	97	106	123	129	136	128	123	105	87	70	67
All-Sky Solar Radiation	RadAvg	RadAvg	1.40	2.28	3.53	4.58	5.38	6.05	6.06	5.18	4.12	2.59	1.57	1.11
	RadStd	RadStd	0.15	0.30	0.37	0.39	0.46	0.33	0.28	0.21	0.39	0.24	0.20	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page